MIL-M-82180A(MC)
1 December 1978
SUPERSEDING
MIL-M-82180(MC)
5 March 1968

MILITARY SPECIFICATION

METER ASSEMBLIES, SKID-MOUNTED: PETROLEUM FUELS

This specification is approved for use by U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers skid-mounted, portable, self-contained meter assemblies.

1.2 Classification. The meter assemblies shall be of the following sizes as specified (see 6.2.1):

Sizes

2-inch (50 to 250 gallons per minute).
3-inch (100 to 650 gallons per minute).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or requests for proposal form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Headquarters Marine Corps, Code LMA-1, Washington, D.C. 20380 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4930

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SPECIFICATIONS

FEDERAL

W-C-440	- Clips, Electrical, General Specification for.
W-R-550	- Rod, Ground (With Attachments).
QQ-S-781	- Strapping, Steel, and Seals.
MMM-A-122	- Adhesive, Butadiene, Acrylonitrile Base, Medium Solids, General Purpose.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-E-52798	- Enamel, Alkyd, Camouflage.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

U.S. MARINE CORPS

DL77C5208A0025	- Meter Assembly, 2-Inch.
DL77C5208A0035	- Meter Assembly, 3-Inch.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

ASTM B 173 - Rope-Lay-Stranded Copper Conductors Having Concentric-Stranded Members, for Electrical Conductors.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE

Hydraulic Institute Standards.

(Application for copies should be addressed to the Hydraulic Institute, 2130 Keith Building, Cleveland, OH 44115.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4 and 6.3).

3.2 Material. Material shall be as specified herein and as shown on the drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3 Design and construction. The meter assemblies shall be in accordance with Data List DL77C5208A0025 for 2-inch and Data List DL77C5208A-0035 for 3-inch and as specified herein. These drawings are end product drawings and no deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the meter assemblies. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.3.1 Ground rod. Each 3-inch meter assembly shall be furnished with a ground rod conforming to W-R-550, Type III, Class B, 5/8 inch in diameter except that, three 2-foot plus or minus 1/16-inch long sections, three couplings, and one driving stud shall be furnished in lieu of the specified 3-foot sections, clamps, ground cable, and ground terminal.

3.3.2 Ground wire assembly. Each 3-inch meter assembly shall be provided with a ground wire assembly conforming to Figure 1.

3.3.3 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.3.4 Castings and forgings. Cast and forged components of the meter assemblies shall be sound and free from gas holes, hard spots, porosity, and cracks. Castings shall not be repaired, plugged, impregnated, brazed, or burned. All fins and roughness shall be removed.

3.3.5 Metal fabrication. Metal used in the fabrication of the meter assemblies shall be free from kinks, ripples, and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Methods of cutting that utilize heat shall not be used for aluminum. Sheared, chipped, and sawed edges shall be free from burrs and slivers. All bends shall be made by controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. Bolt holes shall be accurately formed and have burrs removed. External surfaces shall be free from sharp edges and corners except where sharp edges or corners are required or where they are not detrimental to safety.

3.3.6 Threaded connections and fasteners. All threaded connections and fasteners shall show no evidence of cross threading, mutilation, or detrimental or hazardous burrs. The length of screws and bolts installed with nuts shall be such that the exposed portion is a minimum of two threads. Maximum lengths shall be limited by the nearest standard screw length. All threaded fasteners shall be tight and there shall be no relative movement possible between the attached parts.

3.3.7 Cleaning. The meter assemblies shall be cleaned and dried of all cutting oil, grease, residue of tests, dirt, metal chips, and other extraneous material in accordance with MIL-P-116.

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3.4 Performance characteristics. The metering assembly shall have the operating characteristics specified on Drawing Nos. 77C5208A0031 and 77C5208A0034 (see 4.6.2.2 and 4.6.2.3).

3.4.1 Rough handling. The meter assembly shall be capable of being lifted and dropped without sustaining cracks, stress damage, or permanent deformation (see 4.6.2.4).

3.5 Treatment and painting. All exterior surfaces shall be cleaned, treated, and painted in accordance with MIL-T-704, type B, except that finish coat shall be in accordance with MIL-E-52798 (forest green). In lieu of painting, aluminum surfaces may be anodized in accordance with MIL-A-8625, type optional, class 1. Anodic coatings shall be tested for corrosion and light fastness.

3.6 Identification and instruction plates.

3.6.1 Identification plate. Each meter assembly shall be identified in accordance with MIL-STD-130. The marking shall be applied on plates conforming to MIL-P-514, type I, style 3, composition C, of type optional, grade B, class 1 material. The plates shall be attached to the meter assembly by adhesive, screws, rivets, or bolts in a place where the plate will be visible and legible but will not interfere with the operation and maintenance of the assembly. Adhesive shall conform to MMM-A-122.

3.6.2 Instruction plate. Unless otherwise specified (see 6.2.1), each meter assembly shall be equipped with an instruction plate conforming to MIL-P-514, type III, composition C, of type optional, grade B, class 1 material. The plate shall include all warnings and cautions describing any special or important procedures to be followed in assembling, operating, or servicing the meter assembly. The plate shall include the warning, "NO SMOKING WITHIN 50 FEET." The plate shall be attached to the meter assembly by adhesive, screws, rivets, or bolts in a place where the plate will be visible and legible, but will not interfere with the operation and maintenance of the assembly. Adhesive shall conform to MMM-A-122.

3.7 Workmanship. All parts, components, and assemblies of the meter assemblies shall be free from extraneous material such as sand, dirt, pits, sprues, scale, flux, grease, or oils that would render the meter assemblies unsuitable for their intended use. All parts, components, and assemblies of the meter assemblies shall be free from misalinement, incomplete welds, leaks, or other defects that could impair the operation and safety of the meter assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all

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inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Component and material inspection (see 4.3).
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Packaging inspection (see 4.7).

4.3 Component and material inspection. Components and materials used in the manufacturing, testing, and packaging of the meter assemblies shall be inspected in accordance with the applicable referenced documents. Nonconformance to the applicable referenced documents shall be cause for rejection.

4.4 First article inspection.

4.4.1 Examination. The first article shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. The first article shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined as specified in 4.6.1. Acceptable quality level (AQL) shall be 1.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Individual. Each meter assembly shall be tested as specified in 4.6.2.2. Failure of the test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be tested as specified in 4.6.2.3 and 4.6.2.4 inclusive. AQL shall be 2.5 percent defective.

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4.6 Inspection procedures.

4.6.1 Examination. The meter assembly shall be examined for the following defects:

Major

101. Components missing, damaged, or not in accordance with referenced specifications, standards, and drawings.
102. Coupling cap and plug not attached as specified.
103. Ground rod missing, damaged, or not as specified.
104. Ground wire assembly missing, damaged, or not as specified.
105. Workmanship not as specified.

Minor

201. Treatment and painting not as specified.
202. Identification plate not as specified.
203. Instruction plate not as specified.

4.6.2 Tests.

4.6.2.1 Test fluid. The test fluid shall have a viscosity in the range of 0.7 to 10 centistokes.

4.6.2.2 Hydrostatic pressure. The meter assembly shall be tested as follows:

a. With the outlet end closed with the coupling half cap, connect the inlet end of the meter assembly to test fluid source (see 4.6.2.1) using compatible fittings.
b. Fill the meter assembly with test fluid, using care to eliminate entrapped air.
c. Raise the inlet pressure to not less than 225 pounds force per square inch (lb_f/in^2) and maintain for not less than 5 minutes.
d. During and after the 5-minute period, examine the meter assembly for the leaks and damage.
e. Repeat steps a through d except that the pressure shall be applied to the outlet side of the assembly with the inlet side closed with the coupling half plug.

Nonconformance to 3.4 shall constitute failure of this test.

4.6.2.3 Accuracy and repeatability. The meter assembly shall be tested as follows:

a. The meter assembly shall be tested with the meter and register in a horizontal plane.

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b. Flow rates shall be determined either by quantity meters or rate meters as defined in the test code for centrifugal pumps in the Hydraulic Institute Standards. The test meters used shall have an accuracy of not less than that required for meter assemblies.

c. Calibrate the meter assembly using the test fluid and the meter calibration adjuster so that the registered volume will be within plus or minus 0.2 percent of the true volume. Flow to be used for calibration are as follows:

2-inch -- 148 to 152 gallons per minute.
3-inch -- 598 to 602 gallons per minute.

d. After the meter assembly has been calibrated, test the meter assembly using the test fluid at the following flow rates (plus or minus 1.0 percent):

2-inch -- 50, 125, and 250 gallons per minute.
3-inch -- 100, 125, 350, and 650 gallons per minute.

e. Repeat step d for five consecutive times except that the flow rate shall be 150 gallons per minute (gpm) for 2-inch, and 350 gpm for 3-inch, to determine repeatability.

Nonconformance to 3.4 shall constitute failure of this test.

4.6.2.4 Rough handling. The meter assembly shall be lifted not less than 24 inches and dropped horizontally on a surface consisting of three inches of packed sand over concrete. The meter shall be examined for cracks, breaks, or permanent deformation. Nicks and mars shall not be considered defects. After examination, the meter shall be subjected to the hydrostatic pressure test and the accuracy test. Nonconformance to 3.4.1 shall constitute failure of this test.

4.7 Packaging inspection.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.1.2 Sampling. Sampling for inspection shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective.

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106. Unprotected surfaces not preserved as specified for level A.
107. Technical publications not preserved as specified for level A.
108. Repair parts not preserved as specified for level A.
109. Maintenance tools not preserved in accordance with referenced document for level A.
110. Consolidation not as specified in level A.
111. Exterior containers not as specified for level A.
112. Quantity packed in one exterior container not as specified for level A or B.
113. Strapping not as specified for level A or B.
114. Marking illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Meter assemblies. Unprotected surfaces requiring a preservative shall be preserved in accordance with MIL-P-116.

5.1.1.2 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.1.3 Repair parts. Repair parts shall be preserved in accordance with the applicable criteria of MIL-P-116.

5.1.1.4 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A requirements of PPP-P-40.

5.1.1.5 Consolidation. Each meter and its components shall be placed in a fiberboard box conforming to PPP-B-636 class weather-resistant, grade and variety as required by weight restrictions. The box shall be closed by method V of the box specification appendix.

5.1.2 Commercial. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 2-inch meter assembly. Six 2-inch meter assemblies shall be packed in a box conforming to PPP-B-601, overseas type, grade B or PPP-B-621, class 2. Strapping shall conform to QQ-S-781, class 1, type I or IV,

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size as applicable. Unless otherwise specified (see 6.2.1), strapping shall be finish B. When specified, strapping shall be finish A (see 6.2.1).

5.2.1.2 3-inch meter assembly. Three 3-inch meter assemblies shall be packed in a box as specified in 5.2.1.1.

5.2.2 Level B. The meter assemblies shall be packed as specified in 5.2.1.1 and 5.2.1.2 respectively, except that the boxes shall be domestic type and strapping shall be either finish A or finish B.

5.2.3 Commercial. The meter assemblies shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The meter assemblies are intended for use in volumetric measurements of liquid petroleum fuels. The assemblies are components of an amphibious fuel system that receives, stores, transfers, and dispenses gasoline, diesel fuel, and jet fuel.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. When first article is required for inspection and approval (see 3.1, 4.4, and 6.3).
d. When instruction plate is not required (see 3.6.2).
e. Level of preservation and packing required (see 5.1 and 5.2).
f. When finish A strapping is required (see 5.2.1.1).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a Contract Data Requirements List (DD Form 1423) and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved DD Form 1423 incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

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Paragraph	Date requirement	Applicable DID
4.4	First Article Inspection Procedure.	DI-T-4901
4.4	First Article Inspection Report.	DI-T-4902
4.5	Production/Acceptance Inspection Procedures.	DI-T-4903
4.6	Inspection System Program Plan.	DI-R-4803
4.6	End Item Final Inspection Record.	DI-R-4809

(Copies of data item descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be the first-produced item. The first article should consist of one meter assembly. The contracting officer should include specific instructions of all procurement instruments, regarding arrangements for examination, test, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled or reclaimed material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:

Navy - MC

Project 4930-N215

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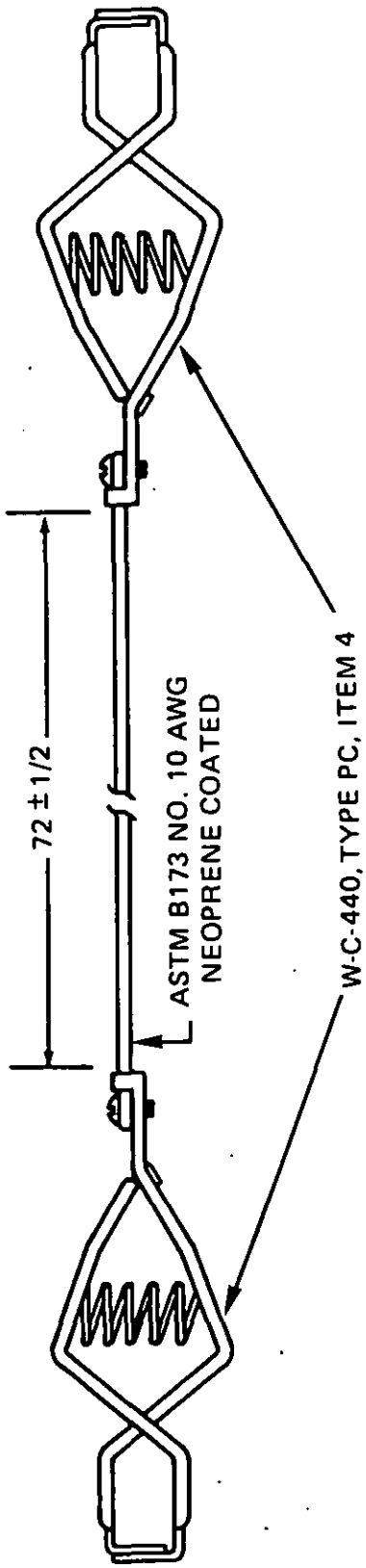


FIGURE 1. Ground wire assembly.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-M-82180A(MC) Meter Assemblies, Skid-Mounted: Petroleum Fuels

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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